Assignment 8 - Big Data Analytics with ThingSpeak

Deliverables: Each student must submit the following:

Presentation Slides (30 points)
- Title slide with student name(s), course, and assignment title.
- Give a clear explanation of each completed task.
- Screenshots or videos of key steps/results.
- Lessons learned and challenges encountered.

Presentation Guidelines
- Use professional formatting (legible fonts, consistent colors, organized layout).
- Ensure clarity: avoid overcrowding slides with text; use bullet points and visuals.
- Provide screenshots or videos as evidence of work completed.
- Keep presentations within the time limit (8–11 minutes).
- Be prepared to answer questions about the process and challenges.

Professionalism & Delivery (10 points)
- Confident and clear presentation delivery
- Ability to answer questions from peers/instructor
- Team collaboration and equal participation

Submission Instructions
- Submit slides (PowerPoint or PDF) to Moodle by the deadline.

Task completion (60 points)

Task 1 — Create ThingSpeak Account

Open ThingSpeak

Open ThingSpeak website: https://thingspeak.mathworks.com/

1. Click:
    - Sign In
2. Create an Thingspeak MathWorks account and Login using MathWorks account

After login, the ThingSpeak dashboard opens.

Task 2 — Create Raw Sensor Data Channel

- **Create New Channel**
    - Click on
        - Channels
        - My Channels

    - New Channel
- **Configure Channel Fields:** Enter

Setting	Value
Name	Factory Sensor Data

Enable these fields:

Field Number	Field Name
Field1	Temperature
Field2	Humidity
Field3	Vibration
Field4	Energy Consumption

- Save the Channel by clicking save channel.

Task 3 — Copy Channel Information

- **Copy Channel ID**

After saving, copy the Channel ID shown at the top.
Example: Channel ID: 1234567
Save it.

- **Copy API Keys**
    - Click: API Keys tab
    - Copy:
        - Read API Key
        - Write API Key

Save them safely.

Task 4 — Create CSV Dataset

- **Create CSV File**
    - Open Notepad or VS Code
    - Create file name: factory_sensor_data.csv

Paste Dataset: Paste EXACTLY this data:

timestamp,temperature,humidity,vibration,energy_consumption

2026-05-01 08:00:00,28.1,62,0.12,120
2026-05-01 08:01:00,28.3,61,0.13,121
2026-05-01 08:02:00,28.5,61,0.14,122
2026-05-01 08:03:00,28.8,60,0.15,123
2026-05-01 08:04:00,29.0,60,0.16,124
2026-05-01 08:05:00,29.2,59,0.17,125
2026-05-01 08:06:00,29.5,59,0.18,126
2026-05-01 08:07:00,29.8,58,0.19,127
2026-05-01 08:08:00,30.0,58,0.20,128
2026-05-01 08:09:00,30.3,57,0.21,129
2026-05-01 08:10:00,30.5,57,0.22,130
2026-05-01 08:11:00,30.8,56,0.23,131
2026-05-01 08:12:00,31.0,56,0.24,132
2026-05-01 08:13:00,31.3,55,0.25,133
2026-05-01 08:14:00,31.5,55,0.26,134
2026-05-01 08:15:00,31.8,54,0.27,135
2026-05-01 08:16:00,32.0,54,0.28,136
2026-05-01 08:17:00,32.3,53,0.29,137
2026-05-01 08:18:00,32.5,53,0.30,138
2026-05-01 08:19:00,32.8,52,0.31,139
2026-05-01 08:20:00,33.0,52,0.32,140
2026-05-01 08:21:00,33.3,51,0.33,141
2026-05-01 08:22:00,33.5,51,0.34,142
2026-05-01 08:23:00,33.8,50,0.35,143
2026-05-01 08:24:00,34.0,50,0.36,144
2026-05-01 08:25:00,34.3,49,0.37,145
2026-05-01 08:26:00,34.5,49,0.38,146
2026-05-01 08:27:00,34.8,48,0.39,147
2026-05-01 08:28:00,35.0,48,0.40,148
2026-05-01 08:29:00,35.3,47,0.41,149
2026-05-01 08:30:00,35.5,47,0.42,150
2026-05-01 08:31:00,35.8,46,0.43,151
2026-05-01 08:32:00,36.0,46,0.44,152
2026-05-01 08:33:00,36.3,45,0.45,153
2026-05-01 08:34:00,36.5,45,0.46,154
2026-05-01 08:35:00,36.8,44,0.47,155
2026-05-01 08:36:00,37.0,44,0.48,156
2026-05-01 08:37:00,37.3,43,0.49,157
2026-05-01 08:38:00,37.5,43,0.50,158
2026-05-01 08:39:00,37.8,42,0.51,159
2026-05-01 08:40:00,38.0,42,0.52,160
2026-05-01 08:41:00,38.3,41,0.53,161
2026-05-01 08:42:00,38.5,41,0.54,162
2026-05-01 08:43:00,38.8,40,0.55,163

2026-05-01 08:44:00,39.0,40,0.56,164
2026-05-01 08:45:00,39.3,39,0.57,165
2026-05-01 08:46:00,39.5,39,0.58,166
2026-05-01 08:47:00,39.8,38,0.59,167
2026-05-01 08:48:00,40.0,38,0.60,168
2026-05-01 08:49:00,40.3,37,0.61,169

Save file as: factory_sensor_data.csv

Task 5 — Upload CSV Dataset

- **Open Import Tool**

Inside your channel:

- Click: Import/Export
    - Import CSV
    - Upload CSV File
        - Choose:
            - `factory_sensor_data.csv`
        - Click:
            - Import

Wait until upload completes.

Task 6 — Verify Uploaded Data

Check Charts: Navigate to Channel you have created then you should now see:

- Temperature graph
- Humidity graph
- Vibration graph
- Energy consumption graph

IMPORTANT: Make sure all 50 rows appear before continuing.

Task 7 — Create Analytics Channel

Create Second Channel

- Click:
    - Channels
    - New Channel

Configure Analytics Fields

Enter:

Setting	Value
Name	Factory Analytics

Enable:

Field Number	Field Name
Field1	Average Temperature
Field2	Maximum Vibration

Click:

- Save Channel

Task 8 — Copy Analytics Channel Information

Copy Analytics Channel ID

Example:

Channel ID: 7654321

Copy Analytics API Keys

Open API Keys

Copy:

- Read API Key
- Write API Key

Task 9 — MATLAB Analysis

Open MATLAB Analysis

Click:

- Apps
- MATLAB Analysis
- New

Paste the following Analysis Code

IMPORTANT: Replace all IDs and Keys with your own values.

```
readChannelID = YOUR_RAW_CHANNEL_ID;
writeChannelID = YOUR_ANALYTICS_CHANNEL_ID;

readAPIKey = 'YOUR_RAW_READ_API_KEY';
writeAPIKey = 'YOUR_ANALYTICS_WRITE_API_KEY';

% Read temperature data
temperature = thingSpeakRead(readChannelID,...
   'Fields',1,...
   'NumPoints',15,...
   'ReadKey',readAPIKey);

% Read vibration data
vibration = thingSpeakRead(readChannelID,...
   'Fields',3,...
```

```
    'NumPoints',15,...
    'ReadKey',readAPIKey);

% Calculate analytics
avgTemp = mean(temperature);
maxVibration = max(vibration);

% Wait to avoid ThingSpeak rate limit
pause(15);

% Write analytics results
thingSpeakWrite(writeChannelID,...
    [avgTemp maxVibration],...
    'Fields',[1 2],...
    'WriteKey',writeAPIKey);
```

Save Analysis

Click:

- Save

Name it: Factory Analytics

Run Analysis

Click on Run

IMPORTANT: After clicking Run:

- Wait at least 20 seconds
- Do NOT click Run repeatedly

This prevents:

Requests are too frequent

Task 10 — Verify Analytics Results

Open Analytics Channel

Go to:

- Factory Analytics channel

You should see:

- Average Temperature value
- Maximum Vibration value

No errors should appear.

Task 11 — MATLAB Visualization

Open MATLAB Visualization

Click:

- Apps
- MATLAB Visualizations
- New

IMPORTANT: Do NOT use MATLAB Analysis for plotting.

Temperature Visualization: Paste the following Visualization Code

```
readChannelID = YOUR_RAW_CHANNEL_ID;

readAPIKey = 'YOUR_RAW_READ_API_KEY';

temperature = thingSpeakRead(readChannelID,...
   'Fields',1,...
   'NumPoints',15,...
   'ReadKey',readAPIKey);

plot(temperature,'-o')

title('Factory Temperature Trend')

xlabel('Data Points')

ylabel('Temperature')

grid on
```

Save Visualization

Click:

- Save

Name: Temperature Visualization

Run Visualization

Click:

- Run

You should now see:

- Temperature trend graph

No errors should appear.

Vibration Visualization:

Create New Visualization: Click on New

Paste the following Vibration Code:

```
readChannelID = YOUR_RAW_CHANNEL_ID;

readAPIKey = 'YOUR_RAW_READ_API_KEY';
```

```
vibration = thingSpeakRead(readChannelID,...
   'Fields',3,...
   'NumPoints',15,...
   'ReadKey',readAPIKey);

plot(vibration,'-r')

title('Machine Vibration Trend')

xlabel('Data Points')

ylabel('Vibration')

grid on
```

Save Visualization

Save as: Vibration Visualization

Run Visualization

Click:

- Run

You should now see:

- Vibration graph